

Total gastrectomy-induced reductions in food intake and weight are counteracted by rikkunshito by attenuating glucagon-like peptide-1 elevation in rats

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Background. Decrease in appetite and weight after total gastrectomy in patients with gastric cancer leads to a decrease in quality of life, increased mortality, and may necessitate discontinuation of adjuvant chemotherapy. The aim of this study is to determine whether rikkunshito, a Japanese herbal medicine, increases food intake and weight after gastrectomy in rats.

Methods. Male rats underwent gastrectomy followed by roux-en-Y reconstruction or sham operation and were then treated with rikkunshito for 14 days starting on postoperative day 3. Daily food intake, weight, plasma glucagon-like peptide-1 (GLP-1), and ghrelin levels were measured. A pilot study to measure pre- and postoperative plasma GLP-1 levels was conducted in patients who underwent total gastrectomy for gastric cancer.

Results. Administration of rikkunshito after gastrectomy in rats significantly increased food intake and weight, which continued for at least 2 weeks after treatment. Both fasting and postprandial plasma GLP-1 levels were increased markedly after gastrectomy compared with sham-operated animals. Increased GLP-1 levels in rats after gastrectomy were suppressed markedly by rikkunshito. rikkunshito had no significant effect on plasma ghrelin levels after gastrectomy. Treatment with a GLP-1 receptor antagonist significantly improved food intake and weight after gastrectomy. Plasma fasting GLP-1 levels in patients with gastric cancer were increased greatly after gastrectomy on postoperative day 1.

Conclusion. Administration of rikkunshito suppresses plasma GLP-1 levels after total gastrectomy, which is associated with recovery from reduced food intake and weight in rats. (Surgery 2016;159:1342-50.)

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WEIGHT LOSS AFTER GASTRECTOMY is an important clinical problem in many patients with gastric cancer. Patients undergoing total gastrectomy experience an average decrease of 15–20% of weight from

baseline,¹⁻³ which may lead to a reduced overall quality of life.⁴⁻⁶ Weight decreases remarkably within the first month after surgery⁷ and usually does not recover, even after 5 years.^{1,3} A study in

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Japanese patients reported that loss of body mass index increases mortality risk as the result of cardiovascular diseases and malignancies.⁸ Weight reduction of 15% or more after gastrectomy in patients with gastric cancer can lead to discontinuation of adjuvant chemotherapy.⁹ Postoperative recovery and maintenance of weight may lead to improved survival.

Weight is regulated by appetite. An increase in appetite is followed by increased caloric intake, which may improve a patient's quality of life after gastrectomy for some patients. Anorexia induced by gastrectomy may be attributable to a loss of stomach retention. It is unknown whether a gastric reservoir can improve food intake.^{10,11} Changes in a number of gut hormone levels after gastrectomy are related to a decrease in appetite. Elevation of cholecystokinin¹² and serotonin¹³ levels and reduction of ghrelin^{14,15} after gastrectomy have been targets for treatment but are not yet clinically useful.

Rikkunshito (RKT), a traditional herbal medicine approved by the Ministry of Health, Labour and Welfare of Japan, is administered orally. RKT is compounded from 8 varieties of crude ingredients: *Atractylodes lanceae* rhizoma (18.6%), *Ginseng radix* (18.6%), *Pinelliae tuber* (18.6%), *Hoelen* (18.6%), *Zizyphi fructus* (9.3%), *Aurantii nobilis pericarpium* (9.3%), *Glycyrrhizae radix* (4.7%), and *Zingiberis rhizoma* (2.3%). RKT improves gastrointestinal symptoms by enhancing gastric emptying.¹⁶ It has been shown recently that RKT reverses decreased levels of ghrelin caused by cancer chemotherapy and thereby restores appetite.¹⁷ RKT is being evaluated as an orexigenic drug. It is unclear whether RKT improves appetite and weight in patients after gastrectomy, where there is no stomach present to produce ghrelin.

The aim of this study is to determine whether RKT increases food intake and weight in rats after gastrectomy. The mechanisms underlying

light/dark cycle (lights on 7:30 AM, off at 7:30 PM) at constant temperature and given an ad libitum conventional diet (CE-2; CLEA Japan, Tokyo, Japan) and water for 1 week. Animals used in this study were all males to avoid sex-related potential biases. All procedures were carried out in a humane manner after we received approval from the Institutional Animal Experiment Committee of Jichi Medical University.

Surgery. Rats were operated at 6 weeks of age, at a weight of approximately 150–210 g. Animals were anesthetized with tribromoethanol, followed by laparotomy. The duodenum and esophagus were divided, and the stomach resected. A roux-en-Y reconstruction was performed in the routine manner, with the jejunum anastomosed in an end-to-side fashion to the esophagus. The jejuno-jejunal anastomosis was performed 4–5 cm from the esophago-jejunal anastomosis in an end-to-side manner. The abdominal wall and skin were closed in layers. For the sham surgery, laparotomy was performed, the stomach and intestine gently manipulated, and the abdomen closed. After surgery, rats were given only water for 3 days. A conventional diet was started on postoperative day 3. Analgesic and antibiotic medications were not used after surgery.

Treatment with RKT. RKT, supplied as a powdered extract (Tsumura & Co. Tokyo, Japan), was suspended in distilled water, adjusted to 1.2% weight per volume, and administered in drinking water for 14 days starting on postoperative day 3 or 17 (Fig 1). For the control groups, distilled water was given ad libitum. Rats were allocated randomly to 4 groups: sham, sham + RKT, gastrectomy, and gastrectomy + RKT.

The rate of recovery of weight due to RKT was calculated by the use of the mean weight of animals in each group according to the following equation:

$$\text{Weight Recovery}(\%) = \frac{(\text{mean weight}(\text{gastrectomy} + \text{RKT}) - \text{mean weight}(\text{gastrectomy}))}{\text{mean weight}(\text{sham}) - \text{mean weight}(\text{gastrectomy})} \times 100$$

decreased feeding and weight after gastrectomy and their recovery by the administration of RKT are also investigated.

METHODS

Animals. Five-week-old male Wistar rats (CLEA Japan, Tokyo, Japan) were maintained with a 12-h

Plasma collection. Six-week-old rats were anesthetized with tribromoethanol and underwent laparotomy. Blood was then sampled directly from the inferior vena cava. After gastrectomy, animals underwent blood sampling from the tail vein on postoperative day 6 and from the vena cava (by laparotomy) on day 10. Blood sampling was done

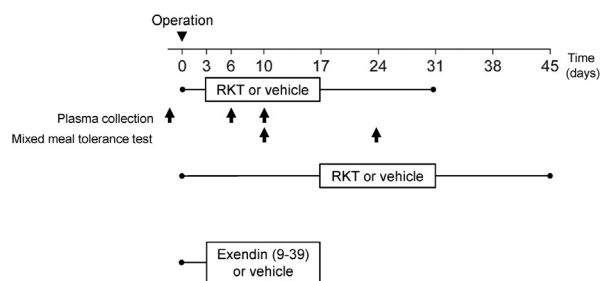


Fig 1. Flow chart showing the overall experimental protocol.

during the light period of the daily cycle (3:00–4:00 PM). Food was not deprived before sampling.

Mixed meal tolerance test. On postoperative days 10 and 24, overnight fasted rats underwent gavage with Ensure H (66% carbohydrate, 17% protein, 17% fat; Abbott Japan, Tokyo, Japan) with 0.5 mL (0.75 kcal)/100 g weight. Blood was collected from the tail veins at 0 (before gavage), 30, 60, and 120 minutes after gavage. Blood glucose was measured with a glucose meter.

Measurement of active GLP-1, acylated ghrelin, insulin. Plasma for measuring active glucagon-like peptide-1 (GLP-1 (7-36)) was added to dipeptidyl peptidase-IV inhibitor (vildagliptin). Plasma for measuring acylated ghrelin was added to HCl. Plasma-active GLP-1 was measured with a GLP-1 (active) enzyme-linked immunosorbent assay kit (AKMGP-011; Shibayagi, Gunma, Japan). Plasma-acylated ghrelin was measured by Active Ghrelin ELISA (Mitsubishi Kagaku Iatron, Tokyo, Japan). Plasma insulin was measured by enzyme-linked immunosorbent assay (Morinaga Institute of Biological Science, Yokohama, Japan).

Subcutaneous exendin (9–39) infusion. The effects of exendin (9–39) (Abgent, San Diego, CA), a specific GLP-1 receptor antagonist, were evaluated in this study. Three days after surgery but before resuming feeding, rats were allocated randomly to 4 groups: sham, sham + exendin (9–39), gastrectomy, gastrectomy + exendin (9–39). Mini-pumps (Alzet mini-osmotic pump model 2002; Durect Corporation, Cupertino, CA) were filled with either exendin (9–39) (0.5 mg for the sham group and 0.4 mg for the gastrectomy group) dissolved in distilled water, or distilled water alone as a control vehicle, which released the solution for 14 days nominally. The pumps were surgically placed in a posterior subcutaneous pocket under general anesthesia. The dose of exendin (9–39) was based on the study by Dailey et al.¹⁸ reporting the dose needed to affect food intake and weight in rats.

Patient study. Patients who underwent gastrectomy for gastric cancer at Jichi Medical University

Hospital from 2013 through 2014 were enrolled in this study if they met the following criteria: (1) curative operation for gastric cancer, (2) total gastrectomy, (3) not treated with preoperative chemotherapy, (4) age 20–80 years, (5) Eastern Cooperative Oncology Group Performance Status 0–1, (6) normal function of organ systems, and (7) written informed consent. Patients were excluded for: (1) reduced oral intake, (2) synchronous or metachronous cancer, (3) hemorrhagic tendency, (4) uncontrolled diabetes mellitus, (5) systemic treatment with corticosteroids, (6) unstable angina or myocardial infarction within 6 months, (7) pulmonary disease treated with continuous oxygen, and (8) pregnancy, breast-feeding, or contemplating pregnancy.

Five patients were enrolled in the study. All patients were fasted the evening before surgery. Blood samples were collected the morning of surgery and on postoperative day 1. Plasma total GLP-1 was measured by Glucagon-Like Peptide-1 total ELISA (EMD Millipore Corporation, Billerica, MA). All patients started oral intake at postoperative days 3–4. The study was approved by the Jichi Medical University Ethics Committee (permit no. A14-052) and registered on UMIN.ac.jp (clinical trial number: UMIN000010568).

Statistical analysis. Data are represented as mean $\pm$ standard error of the mean. Statistical analysis was performed by paired or unpaired *t* test, 1-way and repeated measures 2-way analysis of variance with Tukey-Kramer multiple-comparison test using Prism 6 (GraphPad Software, La Jolla, CA).

RESULTS

Administration of RKT increases food intake and weight after gastrectomy. Cumulative food intake was measured for 21 days and weight for 28 days after surgery. Food intake and weight were decreased markedly in rats after gastrectomy compared with sham-operated rats (Fig 2, A and B) ($P < .05$). RKT (1.2 % weight/volume) was administered in the drinking water for 14 days starting on postoperative day 3. The administration of RKT increased food intake and weight in rats after gastrectomy markedly ($P < .05$), whereas in the sham-operated groups, RKT had no effect on food intake or weight (Fig 2, A and B). On postoperative day 17 when RKT treatment was completed, and 2 weeks after treatment (day 31), weight recovery was 34.3% and 34.8%, respectively, which indicates that the effect of RKT on weight recovery continued after the termination of RKT for at least 2 weeks. RKT, when administered starting on postoperative day 17 for 14 days, resulted in no change in food intake or weight in groups

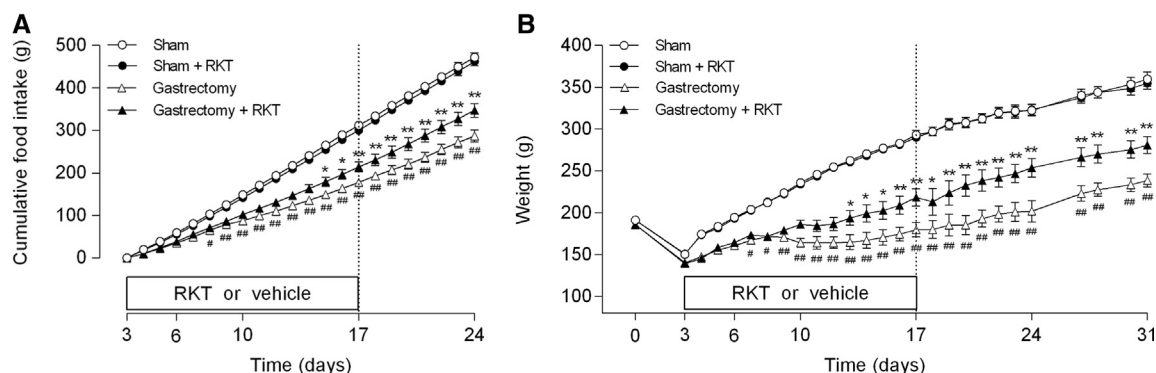


Fig 2. RKT increases food intake and weight markedly after gastrectomy in rats. RKT (1.2 % weight/volume) or distilled water was given for 14 days starting on postoperative day 3. Cumulative food intake (A) and weight (B) decreased markedly in rats after gastrectomy compared with sham-operated rats, whereas administration of RKT increased food intake and weight markedly after gastrectomy. There was no rebound phenomenon after RKT was stopped. For all panels $n = 4$ sham, 4 sham + RKT, 9 gastrectomy, 8 gastrectomy + RKT. Tukey-Kramer's post-test; * $P < .05$, ** $P < .01$, gastrectomy versus gastrectomy + RKT; # $P < .05$, ## $P < .01$, sham versus gastrectomy.

that underwent gastrectomy (Supplementary Fig 1, A and B). These results indicate that the administration of RKT during the early postoperative period markedly restores food intake and weight in rats after gastrectomy and that the effect of RKT continues after treatment.

RKT suppresses GLP-1 elevation in rats during the early postoperative period after gastrectomy. To clarify the mechanism of RKT action, plasma was collected from rats fed ad libitum and hormone levels measured on postoperative days 6 and 10. Plasma active GLP-1 levels tended to increase at day 6 (12.1 ± 1.3 pg/mL) and day 10 (14.1 ± 1.6 pg/mL) after gastrectomy compared with preoperative levels (9.2 ± 0.5 pg/mL), although the differences are not statistically significant. RKT-treated rats showed almost no change at day 6 and day 10 after gastrectomy compared to rats before operation (Fig 3, A). At day 10, plasma-active GLP-1 levels in rats treated with RKT were decreased markedly compared with rats after gastrectomy ($P < .05$) (Fig 3, A). Plasma-acylated ghrelin decreased in rats after gastrectomy compared with rats before operation, with or without RKT, at days 6 and 10. There were no significant differences between rats after gastrectomy and rats after gastrectomy treated with RKT (Fig 3, B). These data indicate that RKT suppresses plasma active GLP-1 elevation after gastrectomy in ad libitum-fed rats but does not affect plasma ghrelin levels.

We conducted a mixed-meal test to further clarify the effect of RKT on plasma-active GLP-1 levels. At postoperative day 10, 7 days after the beginning of RKT treatment, plasma active GLP-1 levels in rats after gastrectomy was greater than in sham-operated rats in a fasted condition (0 min) ($P < .01$), whereas

rats after gastrectomy treated with RKT was lesser than rats after gastrectomy ($P < .05$) (Fig 4, A). After the meal, plasma-active GLP-1 levels in rats after gastrectomy only increased slightly, whereas levels in rats after gastrectomy treated with RKT were stable and remained significantly low compared with rats after gastrectomy ($P < .01$) (Fig 4, A). In the sham-operated groups, there was no change regardless of having received or not received RKT (Fig 4, A). In area under the curve (AUC) analysis of plasma active GLP-1, the level in rats after gastrectomy was greater than the other 3 groups ($P < .01$) (Fig 4, B).

On postoperative day 24, 7 days after RKT treatment was completed, plasma-active GLP-1 levels and their AUC values in a mixed meal tolerance test showed no clinically relevant differences among the 4 groups (Supplementary Fig 2, A and B). Rats after gastrectomy treated with RKT had almost the same AUC-active GLP-1 level compared with the level at day 10 (Fig 4, B and Supplementary Fig 2, B). There were no substantial changes in plasma insulin or blood glucose AUC levels for all groups on day 10 (Supplementary Fig 2, C and D). These data indicate that both fasting and postprandial plasma-active GLP-1 levels are increased after gastrectomy in the early postoperative period, and RKT strongly suppresses elevation of plasma active GLP-1 levels.

Blockade of the GLP-1 receptor after gastrectomy increases food intake and weight. To investigate the influence of GLP-1 on food intake and weight in the early postoperative period, rats underwent continuous subcutaneous infusion of the GLP-1 receptor antagonist exendin (9–39), by mini-pump from postoperative day 3. As a control, distilled water alone was infused. Cumulative food

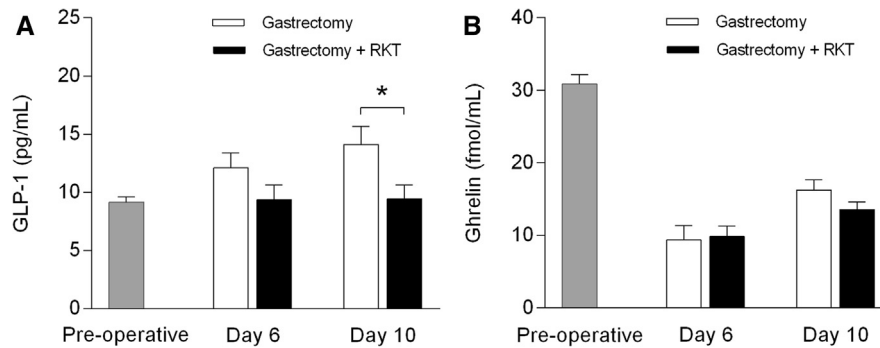


Fig 3. RKT suppresses elevation of plasma GLP-1 levels after gastrectomy in ad libitum-fed rats. Plasma was collected on postoperative days 6 and 10. Gray bars represent the levels in rats before operation. (A) Plasma-active GLP-1 was significantly decreased in rats after gastrectomy that were given RKT, relative to rats after gastrectomy alone on day 10. (B) There was no significant effect of RKT on plasma-acylated ghrelin levels in rats after gastrectomy. For all panels $n = 6$ gastrectomy, 6 gastrectomy + RKT, and 6 rats before operation. Unpaired t test; $*P < .05$.

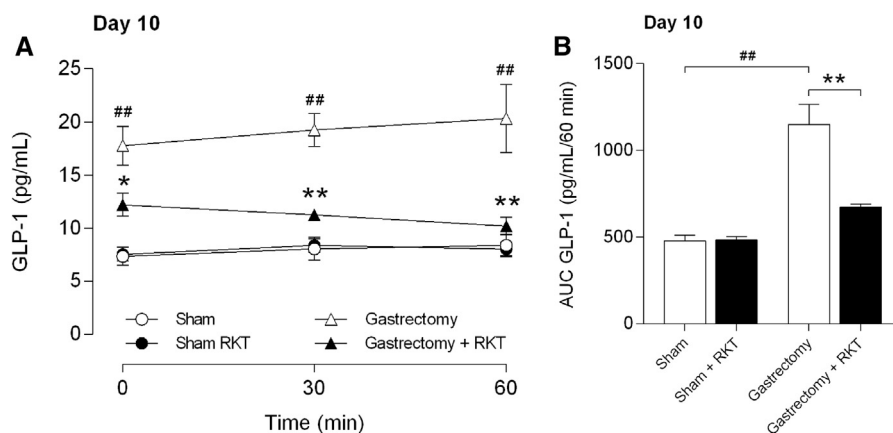


Fig 4. RKT suppresses fasting as well as postprandial GLP-1 elevation after gastrectomy in rats. Plasma-active GLP-1 on postoperative days 10 (7 days after the beginning of RKT treatment) was measured in rats after gastrectomy or sham-operation before (0 minutes) and after gavage feeding of Ensure H. (A) Plasma active GLP-1 levels are increased markedly after gastrectomy compared with sham-operated animals before and after the meal. RKT substantially suppresses fasting as well as postprandial active GLP-1 levels after gastrectomy. (B) AUC analysis of active GLP-1 increased in rats after gastrectomy and is suppressed markedly by RKT. For all panels $n = 4$ sham, 5 sham + RKT, 8 gastrectomy, and 8 gastrectomy + RKT. Tukey-Kramer's post-test; $*P < .05$, $**P < .01$, gastrectomy versus gastrectomy + RKT; $\#P < .05$, $\#\#P < .01$, sham versus gastrectomy.

intake and weight were measured for 9 days after surgery. Animals treated with exendin (9–39) had markedly increased food intake and weight after gastrectomy, whereas there was no effect in sham-operated animals (Fig 5, A and B). These data suggest that increased GLP-1 action after gastrectomy is partly responsible for the observed reduction in food intake and weight loss.

Plasma GLP-1 levels are increased after gastrectomy in patients with gastric cancer. We conducted a pilot clinical study to determine whether human plasma GLP-1 levels are increased after gastrectomy, similar to that observed in rats. There were 5 patients (4 men, 1 woman) with a mean age of

62 years (± 2.4). Their mean weight was 57.6 kg (± 2.8) and mean body mass index 21.9 kg/m² (± 0.8). Three patients underwent laparoscopic resection, and 2 underwent open resection. There were 3 patients with Stage IA tumors, 1 with Stage IB and 1 with Stage IIIA. Blood samples were collected from fasted patients in the morning before and on the first postoperative day after gastrectomy, and plasma total GLP-1 was measured.

In individual data, there was modest variation before and moderate variation of total GLP-1 levels after gastrectomy. All patients showed plasma total GLP-1 elevation after gastrectomy (Fig 6, A), with a clinically relevant difference ($P < .05$) (Fig 6, B).

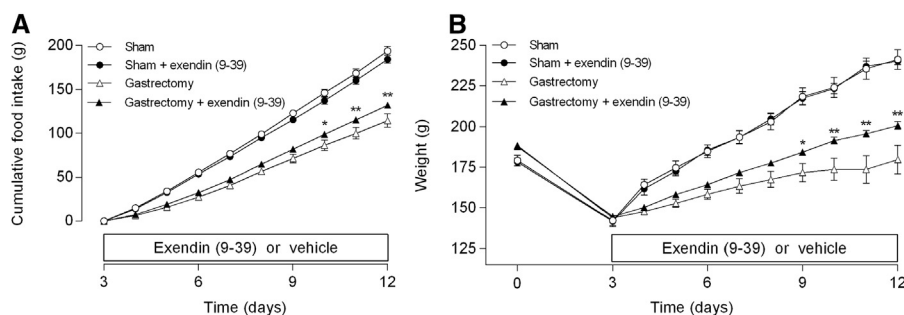


Fig 5. Blockade of GLP-1 receptor after gastrectomy increases food intake and weight. Exendin (9–39) (0.5 mg for the sham group and 0.4 mg for the gastrectomy group) or distilled water was infused subcutaneously over 14 days starting on postoperative day 3. Cumulative food intake (A) and weight (B) increased markedly after gastrectomy when treated with exendin (9–39), compared with gastrectomy alone. For all panels $n = 4$ sham, 4 sham + exendin (9–39), 11 gastrectomy, and 13 gastrectomy + exendin (9–39). Tukey-Kramer's post-test; $*P < .05$, $**P < .01$, gastrectomy versus gastrectomy + exendin (9–39).

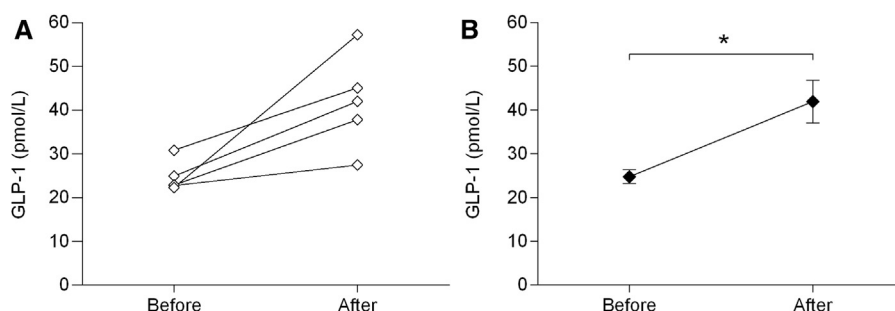


Fig 6. Plasma GLP-1 levels are increased after gastrectomy in patients with gastric cancer. Blood was collected from fasted patients ($n = 5$) the morning before and on postoperative day 1 after gastrectomy. (A) Individual data show a modest variance in total GLP-1 levels before surgery and all patients' total GLP-1 levels are increased on day 1. (B) Plasma total GLP-1 increased markedly on postoperative day 1. Paired t test; $*P < .05$.

This finding indicates that the findings in rats regarding plasma GLP-1 levels after gastrectomy may possibly be translated to human plasma GLP-1 levels.

DISCUSSION

The administration of RKT results in increased food intake and weight when administered in the early postoperative period, starting 3 days after gastrectomy. A mixed-meal test performed on postoperative day 10 shows that plasma GLP-1 levels are increased not only after a meal but also in the fasted state, and RKT suppresses both GLP-1 elevations. Blockade of GLP-1 receptors with exendin (9–39) starting on postoperative day 3 improves food intake and weight after gastrectomy, which mimics the effect of RKT. These results indicate that RKT given in the early postoperative period after gastrectomy leads to increased food intake and weight by counteracting elevated plasma GLP-1 levels. On termination of RKT administration after 2 weeks, increased food intake

and weight are maintained without a rebound effect, which indicates that early postoperative intervention with RKT against decreased food intake and weight loss may affect long-term outcomes. Plasma fasting GLP-1 levels in patients after gastrectomy are increased, suggesting that the findings in rats may apply to humans.

Increased GLP-1 levels after gastrectomy were blunted by postoperative day 24, with no significant difference compared with sham-operated rats. The reason for this finding is unclear, but compensation may occur that normalizes circulating GLP-1 levels in the late postoperative period. RKT had no effect when administered starting on postoperative day 17, which may be attributable to the fact that in the late postoperative period, circulating GLP-1 levels, which is the main target of RKT, are no longer increased.

It is reported that RKT increases plasma orexigenic ghrelin levels in healthy humans and mice¹⁹; however, in the current study, RKT had no effect on acylated ghrelin levels in rats after gastrectomy.

We found that RKT suppresses plasma active GLP-1 elevation early after gastrectomy. GLP-1 is a representative anorexic gut hormone,²⁰⁻²³ which decreases appetite, calorie intake, and weight in humans.^{24,25} To clarify whether increased circulating GLP-1 levels after gastrectomy affects food intake and weight, exendin (9-39) was administered, which is a specific antagonist of the GLP-1 receptor. Food intake and weight in rats after gastrectomy that were treated with exendin (9-39) were increased markedly compared with rats after gastrectomy alone. There was no effect of exendin (9-39) in sham-operated rats. These results indicate that circulating GLP-1 elevation in early after gastrectomy may play a role in the observed decrease in food intake and weight.

GLP-1 is secreted by intestinal endocrine L cells, located in the distal ileum, in response to stimulation by a meal.^{20-23,26} In this study, both fasting and postprandial GLP-1 levels were increased markedly after gastrectomy relative to sham-operated animals in the early postoperative period. Furthermore, fasting GLP-1 levels in humans on postoperative day 1 increased after gastrectomy. The mechanism of GLP-1 elevation after gastrectomy remains unknown. Studies have been performed in patients after intestinal bypass bariatric surgery. The rerouting of food in the intestinal tract results in increased delivery of undigested and unabsorbed nutrients to the distal intestine, overstimulating L cells, with resulting greater levels of GLP-1.^{27,28} The same mechanism may apply to animals after gastrectomy; however, this may not adequately explain the increase in fasting GLP-1 levels. Possibly, other factors, such as removal of the stomach, may continuously stimulate the L cells regardless of the presence of food. Intravenous infusions of GLP-1 decrease fasting hunger in a dose-dependent manner,²⁴ and GLP-1 analogues at high doses induce nausea.²⁹ Thus, fasting, as well as postprandial, endogenous GLP-1 may be involved with appetite, calorie intake, and weight.

RKT did not affect circulating GLP-1 levels in sham-operated rats, suggesting that RKT does not have a direct effect on L cells but modulates the signals specifically after gastrectomy. RKT not only suppresses GLP-1 secretion but may also suppress other gut hormones secreted by L cells, such as GLP-2 or peptide YY. GLP-2 has been suggested to be involved in appetite loss.³⁰⁻³² Peptide YY suppresses appetite, resulting in loss of weight.³³⁻³⁵ RKT may modulate multiple gut hormones as well as GLP-1, increasing appetite resulting in less weight loss after gastrectomy. Further studies are

indicated to elucidate other effects of RKT administration.

We performed a pilot clinical study to investigate whether the findings in rats can translate to humans. Total GLP-1 levels were increased markedly on postoperative day 1 in all 5 patients with gastric cancer who underwent total gastrectomy. Total GLP-1 levels were measured in the fasting state, which is consistent with the results derived from the animal experiments. This finding suggests that circulating GLP-1 levels in the early postoperative period may be involved with appetite and weight loss in patients.

Early oral feeding after gastrectomy is safe and feasible^{36,37} and may decrease postoperative complications.³⁷ Recently, oral intake after surgery is begun earlier.^{37,38} In line with this, the administration of RKT in patients early after surgery, may cause a beneficial effect on appetite and weight loss. Experiments in animals show that the effect of the administration of RKT continues after the treatment stops, and this also should be investigated in patients.

The current study used young male rats. Whether these results also apply to older rats or to female rats needs further investigation. It is also acknowledged that the present study does not determine the possible benefits with longer follow-up, which will require additional studies. In addition, whether the observed orexigenic effects are produced by one of the individual components of RKT or only by the mixture of the ingredients remains to be clarified. In general, the combination of ingredients in RKT has been considered to exert synergistic effects.^{39,40} In this pilot clinical study, the sample size is small, and it is unclear at this time how GLP-1 fluctuates after postoperative day 1 and how GLP-1 affects appetite and weight in patients. Further studies are needed to clarify the action of GLP-1 in patients after gastrectomy.

In conclusion, fasting as well as postprandial elevation of GLP-1 levels are partly responsible for decreased food intake and weight loss in the early postoperative period in rats after gastrectomy. Early postoperative administration of RKT increases food intake and weight in rats after gastrectomy. RKT exerts these effects, at least in part, by counteracting the elevation of GLP-1 levels. Fasting GLP-1 levels are increased in humans after gastrectomy as well. On the basis of these results, clinical trials may be indicated to determine the effect of RKT on appetite and weight loss in patients after gastrectomy.

SUPPLEMENTARY DATA

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.surg.2015.12.003>.

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